

Relative-Metric Affine Tail Recurrences

Exact Coefficients, Vanishing-Forcing Obstructions, and a 330-Step Audit

Prime Dynamics Theory Program

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Abstract

RH-134 proved a strongly contractive raw memory-tail recurrence, with old-tail coefficient below 0.003914. We now transport the recent Gram metric and derive the exact scalar coefficients controlling the relative Rayleigh constant. Suppose

$$D' \preceq rO^*DO + F,$$

where $G, G' \succ 0$ are the source and target recent Gramians and O is an orthogonal target-to-source frame map. If $\gamma^2 = \|G^{-1/2}DG^{-1/2}\|$, then

$$(\gamma')^2 \leq \rho\gamma^2 + q,$$

with the sharp coefficients

$$\rho = r \|G^{1/2}O(G')^{-1/2}\|^2, \quad q = \|(G')^{-1/2}F(G')^{-1/2}\|.$$

Thus raw decay is multiplied by a Gram-comparability factor, and absolute forcing is divided by the weakest target metric direction.

Both amplifications are unavoidable. Without a target Gram lower bound one may have $\|F_\varepsilon\| \rightarrow 0$ while $q_\varepsilon \rightarrow \infty$; similarly, an arbitrarily small raw factor need not give $\rho < 1$. For the Young family from RH-134, write $\rho = A(1 + \tau)$ and $q = q_b + B(1 + \tau^{-1})$. A contractive parameter exists exactly when $A < 1$, after which the affine fixed floor $q/(1 - \rho)$ can be optimized.

A 90-decimal audit of all 330 temporal updates has zero recurrence failures. The classification is sharp: 90 zero-tail transitions are trivial, all 24 first births are contractive, and among 216 recurrent nonzero-tail updates only 51 admit any subunit ρ ; 165 do not. The median recurrent metric base is 9.045 and the maximum is 3.03×10^7 , despite the raw 1/512 decay. Every one of the 75 nontrivial contractive cases has optimized fixed floor below 4.70×10^{-6} . The affine mechanism is therefore excellent when the metric gate opens, but the polar frame gauge fails that gate on most recurrent steps.

1 The metric bridge

Let $G, D \succeq 0$ be a source Gram/tail pair with $G \succ 0$, and similarly let G', D' be the target pair. Assume a raw reduced-coordinate recurrence

$$D' \preceq rO^*DO + F, \quad r \geq 0, \quad F \succeq 0,$$

where O is orthogonal. Define the normalized tails

$$A = G^{-1/2}DG^{-1/2}, \quad A' = (G')^{-1/2}D'(G')^{-1/2}.$$

Conjugating the recurrence by $(G')^{-1/2}$ gives

$$A' \preceq rH^*AH + Q,$$

where

$$H = G^{1/2}O(G')^{-1/2}, \quad Q = (G')^{-1/2}F(G')^{-1/2}.$$

This identity separates the source dynamics from the target metric exactly.

Theorem 1 (Sharp relative affine recurrence). *With*

$$\gamma^2 = \|A\|, \quad (\gamma')^2 = \|A'\|,$$

one has

$$(\gamma')^2 \leq \rho\gamma^2 + q,$$

where

$$\rho = r\|H\|^2, \quad q = \|Q\|.$$

Equivalently, $\|H\|^2$ is the least a such that $O^*GO \preceq aG'$, and q is the least scalar with $F \preceq qG'$. The two coefficients are separately sharp.

Proof. Since $0 \preceq A \preceq \gamma^2 I$, one has $H^*AH \preceq \gamma^2 H^*H \preceq \gamma^2 \|H\|^2 I$. Likewise $Q \preceq \|Q\| I$. Taking the largest eigenvalue proves the bound. The equivalent Loewner characterizations are generalized Rayleigh formulas. Sharpness follows by choosing A and Q supported on a top singular vector of H , with all inequalities aligned in that direction [Horn and Johnson, 1991, Bhatia, 1997]. $\square$

The theorem is exactly the RH-123 recurrence, now with source-level formulas for both coefficients. It also explains why the tiny Euclidean forcing norms in RH-134 were not yet decisive.

2 Two sharp metric obstructions

Proposition 2 (Vanishing absolute forcing can diverge relatively). *There are positive target Gramians G'_ε and positive forcings F_ε such that*

$$\|F_\varepsilon\| \longrightarrow 0, \quad \|(G'_\varepsilon)^{-1/2}F_\varepsilon(G'_\varepsilon)^{-1/2}\| \longrightarrow \infty.$$

Proof. Take $G'_\varepsilon = \text{diag}(1, \varepsilon^2)$ and $F_\varepsilon = \text{diag}(0, \varepsilon)$. Then $\|F_\varepsilon\| = \varepsilon$, while the relative forcing norm is $1/\varepsilon$. $\square$

Proposition 3 (Raw contraction need not be relative contraction). *For every $r > 0$ and $M > 0$ there are $G, G' \succ 0$ and orthogonal O for which $r\|G^{1/2}O(G')^{-1/2}\|^2 > M$.*

Proof. Take $G = I$, $G' = \varepsilon I$, and $O = I$. The coefficient is r/ε , which exceeds M for sufficiently small ε . $\square$

Neither obstruction is numerical. They show that a uniform relative recurrence requires a physical comparison of recent Gramians, not only a better tail estimate. Outward validation can certify such a comparison once it exists, but cannot create the missing gap [Kato, 1995, Higham, 2002].

3 Optimizing the Young tradeoff

The RH-134 moving-frame family has, after metric normalization,

$$\rho(\tau) = A(1 + \tau), \quad q(\tau) = q_b + B(1 + \tau^{-1}), \quad \tau > 0.$$

Here A combines memory decay, envelope ratio, and Gram comparison; q_b is the relative boundary-slice forcing; and B is the relative base of frame forcing.

Proposition 4 (Contractive feasibility and fixed-floor objective). *A parameter with $\rho(\tau) < 1$ exists if and only if $A < 1$. On the interval $0 < \tau < A^{-1} - 1$, the eventual scalar floor is bounded by*

$$\Phi(\tau) = \frac{q_b + B(1 + \tau^{-1})}{1 - A(1 + \tau)}.$$

Its minimum gives the sharpest conclusion available from this Young family. If $A \geq 1$, no choice of τ can make the recurrence contractive.

Proof. The infimum of $A(1 + \tau)$ over $\tau > 0$ is A , proving the feasibility criterion. Iterating a scalar affine recurrence with fixed $\rho < 1$ gives the floor $q/(1 - \rho)$; substitution gives Φ . $\square$

This optimization cannot trade forcing against a metric base already above one. That binary gate is the key diagnostic in the finite audit.

4 Ninety-decimal temporal audit

We rebuild all 30 temporal chains of RH-134 and form the reduced Gramians, weighted tails, metric factors, birth forcing, and frame forcing at 90 decimal digits. Every transition is checked directly against Theorem 1. The Young parameter is optimized on a dense logarithmic grid over its contractive interval when that interval exists.

All 330 recurrences pass. Ninety target tails are zero and are trivially feasible. Twenty-four transitions are first births; their old-tail coefficient is zero, and all have subunit optimized fixed floor. The remaining 216 transitions carry a nonzero old tail. Of these, only 51 have $A < 1$, while 165 have $A \geq 1$ and admit no subunit ρ for any $\tau > 0$.

The recurrent metric base ranges from 0.05069 to 3.0306×10^7 , with median 9.04495. The underlying Gram comparison factor reaches $10^{16.326}$. Blocking becomes more frequent at finer scales: the $\sigma = 0.04$ recurrent updates are all blocked, while scales 0.02 and 0.01 retain 24 and 27 contractive recurrent updates, respectively. The right channel has 96 blocked recurrent steps and only 12 feasible ones; the left channel has 69 blocked and 39 feasible.

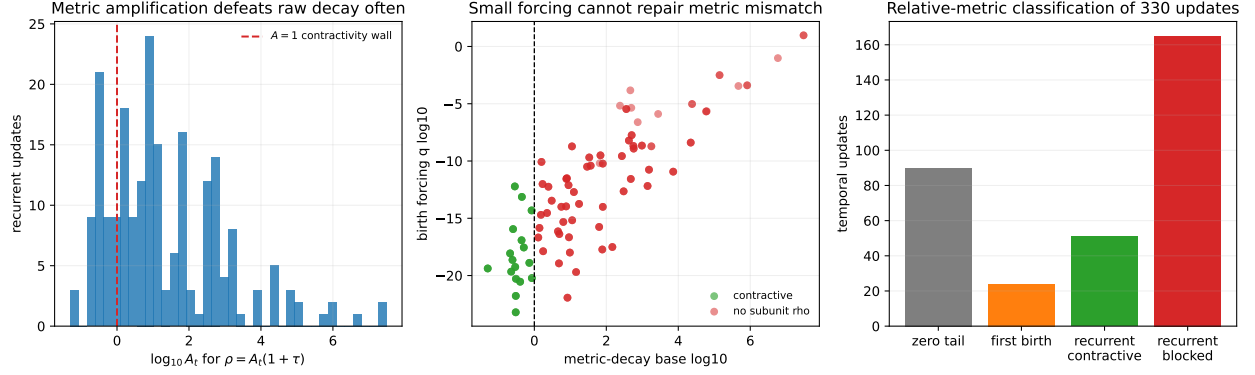


Figure 1: The target-Gram metric factor, not the raw memory decay, is the dominant contractivity gate.

For the 75 nontrivial feasible updates—24 births plus 51 recurrent steps—the optimized fixed floor is uniformly excellent on the frozen data: it lies between 4.35×10^{-25} and 4.70×10^{-6} , with median 3.09×10^{-19} . Therefore forcing size is not the current finite bottleneck once metric contractivity holds. The worst blocked update is the left $\sigma = 0.08$ final transition, where $A \approx 3.03 \times 10^7$ and $q_b \approx 9.406$.

5 Route consequence and boundary

RH-135 closes the algebraic path from the exact memory recursion to the relative affine Rayleigh recurrence. It also gives a decisive negative finite result for the unmodified temporal polar gauge: most recurrent steps are not contractive because recent-Gram metrics change much more violently than the raw memory tail decays.

The next constructive attempt should therefore be metric-aware. One must choose an orthogonal/frame transport that balances two costs: frame defect, which raises forcing, and Gram inflation, which raises ρ . The polar gauge minimizes Euclidean frame displacement but not this affine fixed-floor objective. A metric-balanced Procrustes or generalized polar problem is the natural next gate.

We have not proved uniform subunit ρ , uniform fixed floor, a cross-scale relative recurrence, a positive normalized-base liminf, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

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